

Capstone Proposal Progress Update

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Topic

My capstone will be evaluating the efficacy of using an empirically driven statistical species distribution model as a tool for determining habitat suitability for the eastern oyster (*C. virginica*) within the Pensacola Bay System (PBS) to help drive decision-making in habitat restoration activities. The model will use the Maximum Entropy (MaxEnt) framework, which predicts the relationship between known oyster locations and the underlying environmental conditions using presence-only data.

The resulting suitability grid will be compared to a recent Habitat Suitability Index (HSI) developed for the PBS. HSIs often rely on expert opinion, subjective ranking, and literature derived thresholds for each environmental variable, where as MaxEnt relies on a purely empirically data driven relationship between species occurrence and the environmental conditions. This comparison will test whether the MaxEnt framework can sufficiently produce an efficient alternative to traditional HSI-based restoration planning.

I hypothesize that the MaxEnt approach will result in a highly effective and reproducible habitat suitability workflow that can be updated efficiently as new or more up-to-date data becomes available. The results of the MaxEnt model will be compared to the existing HSI to evaluate model performance and determine potential usefulness for restoration site decision making.

Programming

I anticipate that all data manipulation, processing, and modeling will be done in R using the Positron IDE.

Weeks 4 Progress Update:

(9/8/2026 - 9/14/2026)

Focus

Reviewed literature for environmental factors to include in the species distribution model.

Candidate Variables

The following variables are currently selected based off representation in literature, perceived data availability, or an effort to maintain cross-comparison compatibility with the HSI derived in Geselbracht et al. (2024).

Table 1: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Reason	Reference	Data Source
Salinity	Widely available, large driver of ecosystem suitability	(Peyre et al. 2021; Geselbracht et al. 2024; Pollack et al. 2012; Barneset al. 2007)	Water Quality Portal (DeCicco et al. 2026)
Water temperature	Widely available, large driver of ecosystem suitability	(Peyre et al. 2021; Geselbracht et al. 2024; Pollack et al. 2012; Barneset al. 2007)	Water Quality Portal (DeCicco et al. 2026)
Substrate / bottom type	suitable substrate is required for recruitment	(Peyre et al. 2021; Geselbracht et al. 2024; Barnes et al. 2007)	TBD
Water depth / bathymetry	Depth is a key driver in oyster habitat extent	(Peyre et al. 2021; Geselbracht et al. 2024; Pollack et al. 2012; Barneset al. 2007)	National Centers for Environmental Information (National Centers for Environmental Information 2018)
Dissolved oxygen	Low DO can cause oyster stress and mortality	(Peyre et al. 2021; Geselbracht et al. 2024; Pollack et al. 2012)	Water Quality Portal (DeCicco et al. 2026)
Chlorophyll-a	Common proxy for food resources supporting oyster growth	(Peyre et al. 2021; Geselbracht et al. 2024)	Water Quality Portal (DeCicco et al. 2026)

Table 1: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Reason	Reference	Data Source
Oyster bed locations	important for SDM modeling, which requires species presence data	(Peyre et al. 2021; Geselbracht et al. 2024)	Florida Fish and Wildlife Conservation Commission (Florida Fish and Wildlife Conservation Commission 2015)

Week 5 Progress Update:

(9/15/2026 - 9/21/2026)

Focus

Review literature relating to the underlying logic used in MaxEnt species distribution model.

Maximum entropy modeling of species geographic distributions

Authors: Phillips et al. (2006)

Background

The authors introduce Maxent as a method for species distribution modeling when only presence data are available. The authors note that for many species having both presence and absence data is very unlikely. Maxent uses environmental characteristics of known locations and is able to apply them to unknown locations to predict probability of species occurrence.

Niche-based models can predict the geographic extent but will often overestimate the reality of the species distribution. It is recognized that because an environment might fall within the niche of a given species, doesn't necessarily mean the species are actually established there due to a multitude of geographic, biotic, or anthropogenic barriers. Additionally, there are many assumptions made in such modeling techniques, so the author's express that much care must be taken when choosing which environmental factors to use. With so many assumptions, the authors cite many sources of errors that can be introduced if care is not taken such as using improper temporal or spatial resolutions and scales. Further, the authors state significant errors can also arise from improper data manipulation, poorly optimized environmental models, and interpolation of lower-resolution data.

Finally, the authors acknowledge that there are problems with presence-only data as a whole that must be considered when designing the model. The author's mention things such as

geographic sampling bias, spatial autocorrelation, poor sampling design (e.g. small sample sizes and erroneous sample collection/reporting), and missing or important environmental predictors to name a few.

Maxent

The authors describe the basics of maximum entropy modeling where the goal is to estimate an unknown probability distribution while making as few assumptions as possible. As hinted in the name, the model seeks the distribution that has the maximum entropy (i.e. the most uniform distribution), while maintaining the influence of what is known from the presence data. In species modeling, the model is made up of 3 major parts: 1) the study area pixel grid represents all possible location, 2) known presence observations provide sample locations, and 3) the environmental predictor variables act as features that describe the conditions at the species occurrence locations.

Maxent Advantages:

- Only requires presence data;
- Can use both continuous and categorical data, and incorporate interactions between variables;
- Efficient deterministic algorithms have been developed which are guaranteed to converge on optimal probability distribution;
- The Maxent probability distribution has a well described mathematical definition;
- Over-fitting can be avoided by using regularization techniques;
- Provides flexibility for adding thresholds to continuous predictions;
- Can work on relatively limited data and be applied to all levels of scale.

Maxent Disadvantages:

- At the time of the paper, the Maxent method was still new;
- model guidance was still under development;
- uncertainty estimation was not as well developed when compared to other statistical methods;
- overestimation of environmental variables is possible due to the use of an exponential model for probabilities;
- specialized software requirements

In the paper's methods section, the author's detail the maxent framework. Principally, maxent relies on the assumption that without outside influence, a distribution would become completely uniform, and in the context of species distribution result in a species having the same probability of occurring at any given geographic location. However, species typically occupy an ecological niche which is defined by a series of environmental constraints. As such, Maxent focuses on satisfying each of the constraints (as bounded using known conditions at known locations), while also maximizing the entropy. To do this, the model uses environmental variables as features and tries to use these features to predict a distribution that resembles that of the

observed values at occurrence locations. To prevent overfitting, maxent uses regularization which allows some flexibility between the predicted and observed environmental variables.

Maxent has the ability to transform predictor variables in a number of ways: linear features represent the variable directly (i.e. $f(x) = \textit{Temperature}$), quadratic features use a squared predictor allowing for an exponential change (e.g. $f(x) = \textit{Temperature}^2$), product features allow for variable interactions (e.g. $f(x) = \textit{Temperature} \times \textit{Salinity}$), threshold features allows for predictor values to require certain conditions be met (e.g. $f(x) = \textit{Salinity} > 20$), and binary features which allows predictor variables to represent categorical factors. The ability to transform variables in such away means that maxent has the capability of handling both continuous and categorical variables.

Keyfindings:

In the rest of the paper, the authors describe an implementation of the maxent model and compare it to another more commonly used presence-only modeling method known as a Genetic Algorithm for Rule-Set Prediction (GARP). Using the same predictor data sets and 70/30 training split, both models were able to perform better than random. A threshold-dependent and threshold-independent evaluation was conducted on both models. Threshold-dependent considers the percentage of known points misclassified as unsuitable (omission rate) and the percentage of overall study region that was classified as suitable (predicted area). Threshold-independent receiver operating characteristic (ROC) and area under the ROC curve (AUC) which can describe the overall model performance (e.g. $AUC = 0.5$ roughly equals to random).

The results of the threshold-dependent evaluation showed that both models produced low omission rates. In the threshold-independent test both models AUC values were significantly greater than random, and Maxent generally had higher AUC values than GARP. Overall, maxent proved to be a stronger presence-only modeling approach.

Relevance to capstone:

Phillips et al. (2006) developed the foundational framework of the proposed Maxent species distribution model. They successfully demonstrate using presence-only species data in combination with spatial environmental datasets to predict habitat suitability. They also outline many limitations that should be considered when building the model, selecting predictor variables, and when interpreting model outputs. Finally, the authors provide a useful case study that can help guide my own implementation of the model, describing many useful data manipulation techniques as well as model output evaluation and interpretation guidance.

Next Steps

Review literature on proper spatial modeling techniques to conver environmental point data into continous surface rasters using interpolation techniques such as kriging, etc.

References

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